

MLPR LAB 13

This assignment is to help you familiarize yourself with the installation of libraries required and for the implementation of a simple deep neural network.

INSTRUCTIONS

1. You have to find and load the MNIST digits dataset.
2. Install TensorFlow and Keras libraries.
3. Build your own CNN model using Keras and perform classification on the MNIST dataset.
4. For reference you can follow the following links:
 - <https://www.kaggle.com/code/kanncaa1/convolutional-neural-network-cnn-tutorial>
 - <https://www.askpython.com/python/examples/load-and-plot-mnist-dataset-in-python>
5. Submit the final code with the final performance metrics (Accuracy, Loss, Confusion Matrix, Classification Report) displayed.

REPORT QUESTIONS:

- a) How many layers in the deep neural network were used for features extraction and how many for features classification?
- b) How many kernels were used to extract features on the MNIST images in the first convolutional layer? What were the size of these kernels?
- c) What kind of pooling was done on the data and why?
- d) What were the dimensions of the features set for each layer of the deep CNN? Why were the dimensions reducing in each layer?
- e) How are the outputs of CNN (which are numbers) i.e. activation outputs of the last layer converted to probabilities and 10 classes (for the 10 digits)?

DELIVERABLES: Items to be uploaded on LMS-

- **Main code file along with the complete code, the output displaying all the required final performance metrics and the report answers in one file in PDF format.**
- **Please note: Write your comments along with the code for each step, and show step-by-step outputs as needed. Any plots must be correctly labelled.**
- Your file name should be **yourname_lab13.pdf**. Upload it on the LMS before the due time.